Crop for Trust: Reliability-Aware Industrial Image Classification via ROI Focusing and Generative Annotation Cleaning

KyeongHa Hwang[a] (kyeonghah@gmail.com), SeungHyeok Hong[b,*] (shongdr@gmail.com)

[a] Division of Data Science, The University of Suwon, Hwaseong-si, Gyeonggi-do, Republic of Korea

[b] Division of Social Science & AI, Hankuk University of Foreign Studies, Seoul, Republic of Korea

Corresponding author at: Division of Social Science & AI Convergence, Hankuk University of Foreign Studies, Seoul, Republic of Korea

SeungHyeok Hong

Department of Social Science & AI Convergence, Hankuk University of Foreign Studies, Seoul, Republic of Korea

Tel: +82(0)10 8756-4320

Email: shongdr@gmail.com

Crop for Trust: Reliability-Aware Industrial Image Classification via ROI Focusing and Generative Annotation Cleaning

KyeongHa Hwang[a], SeungHyeok Hong[b,*]

[a]*Department of Data Science, The University of Suwon, Hwaseong-si, Gyeonggi-do, Republic of Korea*
[b]*Department of Social Science & AI Convergence, Hankuk University of Foreign Studies, Seoul, Republic of Korea*

Abstract

Visual biases in "in-the-wild" industrial images—spurious artifacts (e.g., inspection markings) and background clutter—can induce *shortcut learning*, causing models to exploit non-causal cues instead of intrinsic defect patterns. To neutralize these data-level biases, we propose **"Crop for Trust,"** a pre-hoc debiasing framework that sequentially applies (1) **Generative Annotation Cleaning (GAC)** for visual sanitization and (2) **YOLO-based ROI Cropping** for geometric focusing. Unlike traditional inpainting that requires labor-intensive manual masking, GAC leverages instruction-guided generative AI to selectively remove artificial markings in a label-free manner, enabling scalable and cost-efficient industrial deployment. The ROI module then suppresses background bias by restricting learning to defect regions.

Crucially, we show that evaluating models on biased data yields *inflated performance*. Using GAC, we establish a bias-reduced fair baseline and observe a consistent accuracy drop on sanitized data (the bubble-bursting effect), revealing reliance on spurious correlations. Experimental results show that GAC restores intrinsic data reliability by 34% (from 66% to 100%) and that ROI focusing achieves a genuine classification accuracy improvement of **4.69%** over

*Corresponding author.
Email addresses: kyeonghah@gmail.com (KyeongHa Hwang), shongdr@gmail.com (SeungHyeok Hong)

the fair baseline by eliminating background clutter. Grad-CAM visualization confirms the redistribution of attention from spurious cues to defect regions. Overall, Crop for Trust combines instruction-based editing with object-centric focusing to provide an interpretable, data-centric pipeline for trustworthy industrial image classification.

Code and Data Availability: https://huggingface.co/spaces/Crop-Trust-Research/Crop-for-Trust-Demo

Keywords: Industrial Image Classification, Trustworthy AI, Shortcut Learning, Instruction-Guided Generative Editing, Object-Centric ROI Cropping, Reliability-Aware Learning

---

1. Introduction

The success of deep learning-based Visual Inspection Systems in modern industry relies heavily on the availability of large-scale, high-quality datasets. However, data collected in uncontrolled real-world environments ("in-the-wild") inevitably contain various forms of visual bias and contextual artifacts. Researchers often face a dilemma: they must utilize these contaminated datasets to satisfy the *data hunger* of deep neural networks (DNNs). The critical issue is that such data-level biases do more than merely degrade performance; they induce ***shortcut learning***, where the model learns to rely on non-essential features—such as background context or artificial artifacts—rather than actual defect characteristics. This phenomenon poses a structural threat to the **trustworthiness** and **reproducibility** of industrial AI systems.

We specifically address this issue in the context of **Windshield Damage Classification**. Real-world maintenance datasets frequently contain **annotative markings** to highlight damage. A critical issue arises when these markings are spuriously correlated with specific classes (e.g., heavily biased towards the 'Replace' class). Consequently, DNNs may learn to rely on the presence of markings as a facile **visual cue** instead of analyzing minute cracks or break patterns (see Fig 1, Top). Furthermore, since the damage typically occupies a very small

Figure 1: **Visualizing structural biases in industrial data.** The pairs display the original input (left) and the corresponding geometric crop (right). **Top Row:** Annotation bias induced by spurious markings. **Bottom Row:** Background bias caused by environmental clutter.

region of interest (ROI), excessive **background clutter** distracts the model's attention, leading to inconsistent predictions (see Fig 1, Bottom). Thus, even if a model achieves high accuracy, it raises the question: is it making the right decision for the *right reason*, or is it merely exhibiting the *"Clever Hans"* effect driven by biased data?

Existing approaches to mitigate these biases have primarily relied on *post-hoc explainability* techniques to diagnose the issue or *regularization-based training* to suppress it. However, post-hoc analysis is merely a diagnosis, not a cure, and regularization often fails to completely block strong spurious correlations inherent in the input data. That is, structural defects arising at the **input level** cannot be fully resolved by corrections at the **training level**. We reaffirm the principle of *"Garbage In, Garbage Out"*: if the data itself is deceptive, no sophisticated model can make a sound judgment.

To overcome these limitations, we propose **"Crop for Trust,"** a **pre-hoc debiasing framework** designed to fundamentally control visual biases before the training phase. Our approach restores data integrity through a sequential optimization pipeline. First, the **Generative Annotation Cleaning (GAC)** module leverages instruction-guided generative AI to selectively remove class-dependent artifacts. **Unlike traditional inpainting methods that require labor-intensive manual masking for every defect, our approach utilizes semantic instructions to automatically identify and remove artifacts without pixel-level supervision.** This **label-free** capability significantly

enhances scalability and cost-efficiency in industrial MLOps pipelines. Through GAC, we establish a "Fair Baseline" free from artificial cues. Subsequently, **Geometric Focusing via YOLO-based ROI Cropping (YOLO-Crop)** extracts the precise ROI from sanitized images, physically blocking background interference and forcing the model's field of view solely onto the defect.

This study presents the first attempt to combine instruction-guided generative editing with object-centric focusing for interpretable reliability enhancement. The main contributions of this paper are summarized as follows:

- We propose **Crop for Trust**, a unified pre-hoc framework that effectively decouples and mitigates both annotation bias and background clutter without altering the model architecture.
- We empirically demonstrate the **"Bubble Bursting"** effect, where removing spurious markings leads to a performance drop in baseline models, proving their reliance on shortcuts. Conversely, our framework recovers and surpasses this baseline by learning intrinsic features.
- We introduce a quantitative metric, **Intrinsic Data Reliability ($\mathcal{R}_{data}$)**, and provide a reproducible paradigm for securing trustworthy industrial AI systems suitable for label-scarce environments.

2. Related Work

2.1. Visual Bias and Shortcut Learning in Industrial Vision

The phenomenon of "shortcut learning," where models rely on spurious correlations—such as background textures or specific artifacts—instead of causal features, has emerged as a critical bottleneck in computer vision. Zeng et al. (2024) and Birhane & Prabhu (2021) highlight that biases inherent in large-scale datasets inevitably lead to cognitive errors in data-driven models. Similarly, Liang et al. (2023) analyzed the detrimental impact of background context (Background Bias) on object recognition. In the medical domain, which shares structural similarities with industrial inspection regarding data scarcity and

noise, Corbin & Marques (2023) and Gill et al. (2020) have utilized post-hoc explainability tools like Grad-CAM to identify biased attention.

To mitigate these biases, regularization techniques such as **CutMix** Yun et al. (2019) and **MixUp** Zhang et al. (2018) have been widely adopted. These methods aim to prevent overfitting to specific visual cues by mixing samples. Additionally, robust optimization methods like **GroupDRO** Sagawa et al. (2020) explicitly minimize the loss of the worst-performing group to ensure fairness. However, these approaches face practical limitations in industrial settings: augmentation methods effectively mix spurious patterns rather than removing them, and GroupDRO necessitates expensive *spurious-attribute labels* (e.g., manual annotation of 'marked' vs. 'unmarked'), which are rarely available in the wild. **Unlike these training-time mitigations, our framework proposes a pre-hoc approach that structurally eliminates bias at the input level using a label-free generative process, thereby circumventing the need for granular supervision.**

2.2. Region-based Preprocessing for Defect Detection

Research utilizing Regions of Interest (ROI) to mitigate background clutter is a focused area of study. Buongiorno et al. (2023) emphasized the critical role of preprocessing in their survey on industrial surface defect detection, and a recent review by Wu et al. (2025) reported that integrating region-based preprocessing with deep learning is a major trend for improving classification consistency. While the YOLO architecture Redmon et al. (2016) is renowned for real-time object detection, previous studies have primarily utilized it merely as a locator or for simple cropping. This study extends this paradigm by redefining YOLO-Crop not just as a preprocessing step, but as an **active de-biasing mechanism** (Geometric Focusing) that physically filters out environmental noise to rectify the model's visual attention and enhance data reliability.

2.3. Generative AI for Data Augmentation and Cleaning

Generative AI has recently emerged as a key technology for addressing data scarcity. Sampath et al. (2021) analyzed the use of GANs to improve imbal-

anced data distributions, and Aldubaikhi & Patel (2025) discussed generative augmentation as an effective strategy for small-object detection tasks. Recent advancements in semantic inpainting, such as **RePaint** Lugmayr et al. (2022) and text-guided frameworks like **CLIPAway** Man et al. (2024), have demonstrated high-fidelity restoration capabilities. However, most existing research focuses on **"addition"** (i.e., generating missing parts or synthesizing new samples) and typically requires pixel-level masks for targeted removal. In contrast, this study utilizes generative AI as a tool for **"subtraction"** (i.e., cleaning and removing artifacts). Drawing inspiration from Elsharif et al. (2025), who investigated the relationship between generative AI and bias mitigation, we apply this concept to precisely eliminate **annotation bias** in industrial data. This allows us to construct a "Fair Baseline," presenting a novel methodology for data-centric reliability.

3. Problem Formulation: Anatomy of Industrial Data Bias

To reflect the complexities of "in-the-wild" industrial environments, we constructed a dataset utilizing real-world windshield damage images collected from vehicle repair shops. Each sample was rigorously inspected by domain experts and labeled as either **Repair** or **Replace**. The strict classification criteria applied by the experts are defined as follows:

- **Crack:** Length $\leq$ 15 cm (6 inches) $\rightarrow$ Repair; otherwise $\rightarrow$ Replace.
- **Stone Chip, Bulls Eye:** Diameter $<$ 2.5 cm (1 inch) $\rightarrow$ Repair; otherwise $\rightarrow$ Replace.
- **Star Break:** Crack length from impact point $\leq$ 7.5 cm (3 inches) $\rightarrow$ Repair; otherwise $\rightarrow$ Replace.
- **Edge Damage:** Classified as Replace if located at the glass edge, affecting structural integrity.

The dataset was randomly partitioned into Train, Validation, and Test sets with a ratio of 8:1:1. While standard augmentations (e.g., random rotation $\pm 15°$, flip, color jittering) were applied during training to ensure basic generalization, these geometric transformations are insufficient to address the **structural bias** inherent in the data distribution, as analyzed below.

3.1. Quantification of Spurious Correlations: The Class-Dependent Artifact

Table 1: **Statistical quantification of annotation bias.** The table reports the total number of samples per class, with the count of **marked images** shown in parentheses. Note the disproportionate frequency of markings in the 'Replace' class compared to 'Repair'.

Class	**Train**	**Validation**	**Test**
Repair	897 (87)	111 (19)	109 (17)
Replace	942 (514)	57 (52)	56 (50)

We conducted a statistical analysis to quantify the strength of the association between visual artifacts and specific classes. Table 1 presents the distribution of spurious markings across each dataset split.

As shown in the Training set (Table 1), while the class distribution is numerically balanced (Repair: 897 vs. Replace: 942), a critical imbalance exists in the *conditional probability of marking presence*. Only **9.7%** (87/897) of Repair images contain markings, whereas over **54.5%** (514/942) of Replace images do. This discrepancy stems from the field practice of explicitly annotating damages deemed "severe" for easier identification. Consequently, the presence of markings becomes a dominant predictor, creating a significant disparity in posterior probabilities:

$$P(\text{Replace}|\text{Marking}) \gg P(\text{Repair}|\text{Marking}) \tag{1}$$

This imbalance induces strong **spurious correlations**. To verify this, we tested a **"Naive Rule-based Model"** that classifies images solely based on the presence of markings (i.e., if Marked $\rightarrow$ Replace, else $\rightarrow$ Repair), completely

ignoring intrinsic defect features. Notably, this simple heuristic achieved an accuracy of **86.1%** on the Test Set.

This result provides empirical evidence that the high performance of conventional models is likely not attributed to recognizing actual damage patterns, but rather to exploiting the *shortcut* of marking presence. In essence, visual markings function as a **confounder** that artificially *inflates* performance. Without controlling this factor, the **trustworthiness** of the system cannot be guaranteed. This quantitative analysis underscores that our proposed **Pre-hoc Debiasing strategy** is not merely an enhancement, but a structural necessity.

3.2. Attentional Misalignment Induced by Background Clutter

Modern deep learning architectures, despite their capacity, are inherently vulnerable to **background clutter** due to the mechanism of feature extraction. While CNN-based models He et al. (2016) utilize local receptive fields, they struggle to strictly dissociate foreground objects from high-frequency background textures. Similarly, Vision Transformer (ViT) architectures Dosovitskiy et al. (2021), in their pursuit of global context, tend to excessively integrate irrelevant environmental information with defect features.

This issue is exacerbated in our industrial windshield dataset. Quantitative analysis reveals that the actual Region of Interest (ROI) for damage typically occupies an extremely small fraction (estimated at less than 5%) of the total image pixels. Under such conditions of *extreme foreground-background imbalance*, the effective **Signal-to-Noise Ratio (SNR)** of the defect is critically low. Consequently, the model is prone to misinterpreting **dominant background signals**—such as workshop infrastructure, wiper blades, or surrounding text—as primary decision cues, rather than focusing on minute damage patterns. This leads to a phenomenon we term **"Attentional Misalignment,"** where the model's gaze is diverted to non-causal regions.

Recent studies confirm this as a pervasive structural issue. The *"Contextual Correlation Theory"* proposed by Zeng et al. (2024) theoretically supports that statistical dependencies between objects and backgrounds inevitably dis-

tort the model's focus. Furthermore, Liang et al. (2023) imply that background information is not merely noise but a cause of *"context-dependent prediction,"* exerting a direct causal influence on classification results. As will be detailed in Section 5.2, our preliminary Grad-CAM analysis corroborates this vulnerability: baseline models exhibited high activation on strong reflections or background textures rather than on the damage itself. This suggests that preprocessing to physically block the background (i.e., Geometric Focusing) is not a mere option but a prerequisite for securing **attentional integrity**.

3.3. The Scarcity Dilemma: Structural Constraints of Industrial Data

Our dataset comprises 1,325 original images. Even after applying selective augmentation to the minority class to mitigate class imbalance, the total training set remains limited to 2,172 samples. Such a small sample size not only induces overfitting but also serves as a fundamental factor driving the model to rely on easier shortcuts—such as "markings" or "background"—rather than learning complex "damage" features. In essence, data scarcity acts as a **catalyst**, amplifying the model's **bias sensitivity**.

Collecting large-scale data in the industrial domain is practically challenging due to the low frequency of actual defects and uncontrollable capture environments. This *domain-specific data scarcity* is a chronic structural constraint reported across various fields, including medical imaging and semiconductor inspection Elsharif et al. (2025). While transfer learning with pretrained models (e.g., ImageNet, COCO) is a common mitigation strategy, there exists a significant **distributional discrepancy**. Pretrained models are typically optimized for *object-centric* natural images, leading to a **representation mismatch** when transferred to *texture-centric* industrial defects. Ultimately, the scarcity of domain-specific data destabilizes the learning process, making models more susceptible to the aforementioned visual biases.

This study identifies these dual limitations—visual bias and data scarcity—and seeks to simultaneously achieve bias control and reliability enhancement through the **pre-hoc bias mitigation framework** proposed in Section 4.

4. Methodology

This section details **"Crop for Trust,"** a pre-hoc bias mitigation framework designed to address the three structural constraints of industrial data identified in Section 3: (1) class-dependent marking artifacts, (2) extreme foreground-background imbalance, and (3) domain-specific data scarcity.

4.1. Problem Formulation

We formulate the windshield damage classification task as learning a mapping function $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$, where $\mathcal{X}$ denotes the input image space and $\mathcal{Y} = \{\text{Repair}, \text{Replace}\}$ represents the label space. In real-world industrial scenarios, a raw input image $x \in \mathcal{X}$ is modeled as a composite of intrinsic damage features (x_{dmg}), background bias (x_{bg}), and spurious artifacts (x_{art}). Unlike simple additive noise, these components interact via occlusion:

$$x = \mathbf{M} \odot x_{dmg} + (\mathbf{1} - \mathbf{M}) \odot x_{bg} + x_{art} \tag{2}$$

where $\mathbf{M} \in \{0, 1\}^{H \times W}$ denotes the **spatial binary mask** indicating the precise pixel locations of the defect, and $\odot$ represents element-wise multiplication. This formulation explicitly models the **occlusion** phenomenon: intrinsic damage features (x_{dmg}) physically replace the background (x_{bg}) at the defect coordinates, whereas artifacts (x_{art}) are superimposed as spurious cues.

Conventional models often fail by minimizing the loss function through reliance on x_{art} or the background context within $(\mathbf{1} - \mathbf{M}) \odot x_{bg}$ rather than learning x_{dmg}, a phenomenon known as *shortcut learning*. Therefore, our objective is to design a pre-hoc transformation function $\Phi(\cdot) = \Phi_{focus}(\Phi_{clean}(\cdot))$ that suppresses x_{art} and isolates x_{dmg} prior to training.

To achieve this, we introduce a **Sequential Debiasing Framework** composed of two complementary modules:

- **Generative Annotation Cleaning (GAC):** Leverages instruction-guided generative AI to selectively remove x_{art}, restoring *Visual Fairness*.

- **Geometric Focusing (via YOLO-Crop):** Utilizes object detection to geometrically eliminate background bias and extract x_{dmg}, inducing *Attention Focus*.

4.2. Framework Overview

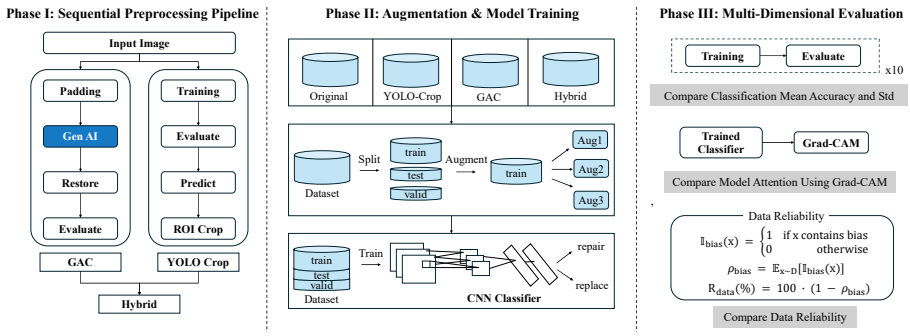


Figure 2: **Architecture of the "Crop for Trust" framework.** The workflow proceeds sequentially: (1) **Preprocessing (Phase I)** creates sanitized and focused inputs via GAC and YOLO; (2) **Training (Phase II)** utilizes four distinct dataset scenarios ($\mathcal{D}_{orig}, \mathcal{D}_{crop}, \mathcal{D}_{GAC}, \mathcal{D}_{hybrid}$) to isolate bias factors; and (3) **Evaluation (Phase III)** validates performance using classification metrics, attention visualization, and the proposed reliability score ($\mathcal{R}_{data}$).

The overall architecture is illustrated in Fig 2. The pipeline operates through three integrated phases.

Phase I: Preprocessing and Dataset Construction. The core of our framework lies in the data preparation stage (Left block of Fig 2). We establish the original dataset ($\mathcal{D}_{orig}$) as a biased baseline and derive three sanitized variants to isolate the impact of each bias factor:

1. **GAC Dataset ($\mathcal{D}_{GAC}$):** A bias-controlled baseline where only marking artifacts (x_{art}) are removed while the background is preserved.
2. **YOLO-Crop Dataset ($\mathcal{D}_{crop}$):** A focused dataset where background noise is removed, retaining only the ROI. Note that marking artifacts within the ROI persist in this set.

3. **Hybrid Dataset ($\mathcal{D}_{hybrid}$):** A fully purified dataset obtained by applying YOLO-Crop to GAC-processed images, eliminating both annotation and background biases.

Phase II & III: Training and Hypothesis Verification. As depicted in the Middle and Right blocks of Fig 2, the constructed datasets are fed into an identical classification backbone to ensure fair comparison. We adopt the compound scaling philosophy Tan & Le (2019) to evaluate robustness across varying model scales (Small to Large) and architectures. The final assessment (Phase III) utilizes a multi-dimensional protocol analyzing classification accuracy, Grad-CAM attention maps, and our proposed Data Reliability metric ($\mathcal{R}_{data}$). Through this rigorous design, we aim to verify two key hypotheses:

- **H1 (Attention Rectification):** Applying geometric focusing (YOLO-Crop) physically removes non-causal background features, thereby rectifying the model's attention toward actual damage regions.
- **H2 (The Bubble Bursting Effect):** Eliminating spurious artifacts (x_{art}) via GAC will reveal the model's reliance on shortcuts. This may cause a drop in nominal accuracy (bubble bursting), which we interpret not as a failure, but as the establishment of a truthful performance baseline.

4.3. Geometric Focusing: Implementation via YOLO-based ROI Cropping

The first module, Geometric Focusing, is implemented using the YOLO architecture to serve as a spatial filter designed to extract the precise Region of Interest (ROI) (x_{dmg}) while physically eliminating irrelevant background bias (x_{bg}).

Mechanism: Implicit Data Augmentation. Beyond simple noise removal, this module functions as an engine for *Implicit Data Augmentation*. Consider a raw image I_i containing k distinct damage instances. We treat each damage as an independent object, defined by the bounding box set $B_i = \{b_1, b_2, \ldots, b_k\}$.

The YOLO-Crop process transforms a single sample (I_i, y_i) into k independent training samples:

$$\mathcal{D}_{crop} = \bigcup_{i=1}^{N} \{(\text{Crop}(I_i, b_j), y_i) \mid b_j \in B_i\} \tag{3}$$

This **instance-level decomposition** effectively decouples multiple defects within a single field of view, statistically amplifying the effective training size and structurally mitigating the data scarcity issue discussed in Section 3.3.

Implementation Details. We fine-tuned various YOLO variants (v2, v4, v5, v8-Series) pre-trained on COCO using our Windshield Damage Dataset. To ensure reproducibility and fair comparison, the training configuration was standardized: **Input Resolution** 416×416; **Epochs** 400; **Batch size** 16; **Optimizer** AdamW; and **Precision** AMP (FP16). The best checkpoint was selected based on Validation mAP_{50-95}.

Structural Limitation and Motivation for GAC. While Geometric Focusing primarily targets the suppression of background clutter (x_{bg}), it effectively serves as a spatial filter that incidentally removes artifacts located outside the ROI (x_{art}^{ext}). However, this removal is merely a byproduct of cropping; the fundamental limitation persists that **marking artifacts located *within* the ROI (x_{art}^{int}) remain untouched.** Since the cropping operation preserves the pixel content inside the bounding box, any artifact overlapping with the defect is carried over to the training data. Consequently, even with the reduction of environmental bias, the risk of **"internal shortcut learning"** via these residual markings remains a critical threat. This structural vulnerability necessitates our second module, **Generative Annotation Cleaning (GAC)**, designed to strictly sanitize the pixel-level content within the focused view.

4.4. Generative Annotation Cleaning (GAC): Concept and Validation

The second core module, **Generative Annotation Cleaning (GAC)**, is designed to selectively eliminate spurious marking artifacts (x_{art}) to establish a "Fair Baseline." This module serves the dual purpose of **Decoupling Bias**

Impact and **Rectifying Visual Imbalance**, forcing the model to learn from intrinsic damage cues rather than artificial indicators.

Instruction-Guided Sanitization. We leverage the multimodal capabilities of **Gemini 2.5 Flash** Google DeepMind (2025) for instruction-guided editing. Conventional inpainting methods (e.g., LAMA, Stable Diffusion) suffer from a **"Mask Dependency Bottleneck"**—they necessitate pre-defined binary masks, requiring either labor-intensive manual annotation or an auxiliary segmentation model. This creates a circular dependency in label-scarce industrial environments. In contrast, our approach utilizes **semantic reasoning** via natural language instructions to automatically identify and remove artifacts. By unifying detection and removal into a single label-free inference process, GAC significantly enhances **scalability** and streamlines the industrial MLOps pipeline.

The process is governed by a structured prompt $\mathcal{P}$ designed to enforce **Dual Constraints of Preservation and Removal**:

> *"Retain the damage area and cracks exactly as they are. Remove all artificial annotations (lines, circles, text) regardless of color or shape. Restore the erased glass surface with a natural, realistic texture."*

Input images are padded to 1024×1024 to prevent aspect ratio distortion during the generative process.

Self-Supervised Evaluation Protocol. Since generative editing carries the inherent risk of **hallucination** (i.e., altering the actual defect structure), verifying the integrity of the generated data is paramount. To this end, we designed a rigorous **Self-Supervised Evaluation Protocol**, as illustrated in Fig. 3. Since ground truth for "cleaned" real images does not exist, we synthesize virtual markings on clean images to create a controlled testbed where the original clean image serves as the absolute Ground Truth.

The procedure, detailed in Algorithm 1, involves three phases:

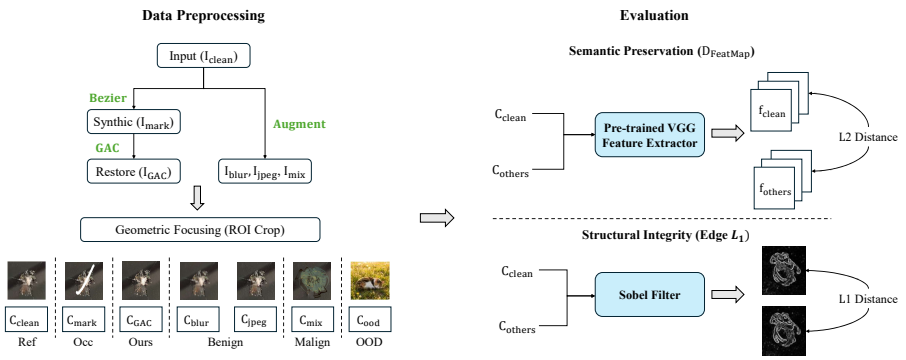


Figure 3: **Validating generative safety via the Synthesis-Restoration protocol.** (Left) Clean samples are artificially corrupted with Bezier curves (I_{mark}) and restored via GAC (I_{GAC}). (Right) The restored ROI crops are quantitatively benchmarked against Safety Anchors (Blur, JPEG) and Failure Anchors (Mix, OOD). Metrics include Semantic Distance ($D_{FeatMap}$) and Structural Integrity (Edge L_1) to detect potential hallucinations.

1. **Synthesis (Hard Constraint):** We synthesize realistic Bezier curve markings (x_{art}^{syn}) that strictly overlap the defect regions in clean images (I_{clean}). This enforces a high-difficulty scenario, compelling the model to hallucinate the occluded structure based on context.
2. **Restoration:** The synthetic I_{mark} is processed by GAC to produce the restored I_{GAC}.
3. **Geometric Focusing:** Quantitative evaluation is performed strictly on the ROI crops (C_{clean}, C_{GAC}) to prevent metric dilution by background noise.

Comparative Anchors and Metrics. To benchmark the safety of GAC, we establish relative safety bounds using four anchor groups:

- **Occluded Baseline (C_{mark}):** Quantifies the initial information loss due to markings.
- **Benign Anchors (Safety Bound):** Gaussian Blur ($\sigma = 1.0$) and JPEG ($Q = 50$), representing acceptable quality degradation where defect identity is preserved.

Algorithm 1 Self-Supervised GAC Evaluation Protocol

Require: Clean dataset $\mathcal{D}_{clean}$, GT boxes $\mathcal{B}_{gt}$

Ensure: Metrics $\mathcal{M} = \{\text{PSNR}, \text{SSIM}, \text{EdgeL1}, D_{FeatMap}\}$

Initialize: Feature Extractor Φ (VGG-16, fixed)
for all $I_{clean} \in \mathcal{D}_{clean}$ **do**
  $b \leftarrow \textsc{GetBox}(\mathcal{B}_{gt}, I_{clean})$
  // Phase 1: Synthesis & Restoration
  $I_{mark} \leftarrow \textsc{Bezier}(I_{clean}, b, \text{overlap} = \text{true})$
  $I_{GAC} \leftarrow \text{GAC}(I_{mark}, \mathcal{P})$
  $I_{anchors} \leftarrow \textsc{Generate}(I_{clean}, \{\textsc{Blur}, \textsc{Mix}, \ldots\})$
  // Phase 2: ROI Extraction
  $\{C_{clean}, C_{GAC}, C_{anchors}\} \leftarrow \text{ROI_}\textsc{Crop}(\{I_{clean}, I_{GAC}, I_{anchors}\}, b)$
  // Phase 3: Metric Calculation
  for all $C_* \in \{C_{GAC}\} \cup C_{anchors}$ **do**
    $D_{FeatMap} \leftarrow \|\Phi(C_{clean}) - \Phi(C_*)\|_2$
    $\text{Metrics} \leftarrow \textsc{StructuralMetrics}(C_{clean}, C_*)$
    $\textsc{Record}(\mathcal{M}, C_*, \text{Metrics})$
  end for
end for
return $\textsc{Aggregate}(\mathcal{M})$

- **Malign Anchor (Failure Bound):** A 'Mix' image (C_{mix}) created via Poisson Blending of heterogeneous defects, representing structural corruption.
- **OOD Anchor:** An out-of-domain image serving as the maximum dissimilarity baseline.

Evaluation utilizes **Semantic Distance ($D_{FeatMap}$)** via VGG-16 feature space and **Structural Integrity (Edge L_1)** via Sobel edge difference, alongside traditional PSNR/SSIM.

4.5. Classification Protocol and Experimental Setup

The final phase of our framework involves applying a unified training pipeline across the four constructed datasets ($\mathcal{D}_{orig}, \mathcal{D}_{GAC}, \mathcal{D}_{crop}, \mathcal{D}_{hybrid}$). This design allows for a quantitative analysis of how the proposed preprocessing strategies influence the model's *Learning Dynamics* and *Generalization* capabilities.

Architectural Agnosticism and Model Selection. To demonstrate that our framework yields universal performance improvements independent of specific model structures, we selected five representative CNN architectures as benchmark backbones. These range from **High-Capacity Models** (VGG16, ResNet152) capable of learning complex features to **Lightweight Models** (MobileNet v1, v2, v3) designed for constrained computational resources.

- *Why CNNs over ViTs?* While Vision Transformers (ViT) have recently gained traction, we deliberately prioritized mature CNN architectures. This decision reflects the rigid practical constraints of industrial inspection lines, where **inference latency** and **edge hardware compatibility** are paramount. Consequently, CNNs remain the *de facto* standard for real-time deployment in these resource-constrained environments, making them the most relevant baseline for validating the practical utility of our framework.

All models were initialized with ImageNet pre-trained weights and fine-tuned on our target domain.

Standardized Training Protocol. To ensure the reliability of results and fair comparison, all experimental scenarios adhered to a strictly controlled hyperparameter setting. We utilized the **AdamW** optimizer with an initial learning rate of 10^{-3}, a **Batch Size** of 32, and the standard **Cross-Entropy Loss**. The training budget was set to 80 epochs per trial, with the best checkpoint selected based on the minimum validation loss.

Reliability Protocol (Statistical Significance). To mitigate the randomness inherent in deep learning training, we implemented a rigorous reliability protocol. Each experiment was repeated independently **10 times** under random seed initialization. Final performance metrics are reported as the **Mean Accuracy** and **Standard Deviation**. This experimental design aims to multi-dimensionally elucidate how the combination of *Visual Sanitization* (via GAC) and *Spatial Focusing* (via YOLO-Crop) enhances both the model's **Decision Stability** and **Attention Alignment**.

5. Experiments and Results

In this section, we empirically validate the effectiveness of the proposed 'Crop for Trust' framework through a three-phase experimental protocol. First, we verify the safety and purification quality of the GAC module (Section 5.1). Second, we qualitatively analyze the rectification of model attention using Grad-CAM (Section 5.2). Finally, we evaluate the impact on data reliability and classification performance against various baselines (Sections 5.3-5.7).

5.1. Validation of Generative Sanitization: Preservation vs. Removal

Table 2: **Quantitative benchmarking of restoration quality.** The proposed GAC is compared against safety bounds (Blur, JPEG) and failure bounds (Mix). The metrics confirm that GAC achieves high structural fidelity (SSIM > 0.99) and semantic preservation ($D_{FeatMap} \approx 0.01$), closely aligning with benign deformations.

Method	**PSNR** $\uparrow$	**SSIM** $\uparrow$	**Edge** $L_1 \downarrow$	$D_{FeatMap} \downarrow$
Marked (C_{mark})	41.1919	0.9840	14.4813	0.0057
BLUR	37.7935	0.9998	12.9871	0.0060
GAC (Ours)	**35.0135**	**0.9954**	**15.7693**	**0.0112**
JPEG	37.3273	0.9986	9.2649	0.0082
MIX (Malign)	30.8957	0.9344	45.4359	0.0230
OOD	27.8941	0.6717	131.2252	0.0455

The quantitative evaluation based on the self-supervised protocol (Algorithm 1) is summarized in Table 2, and the safety verification via scatter plot analysis is illustrated in Fig. 4.

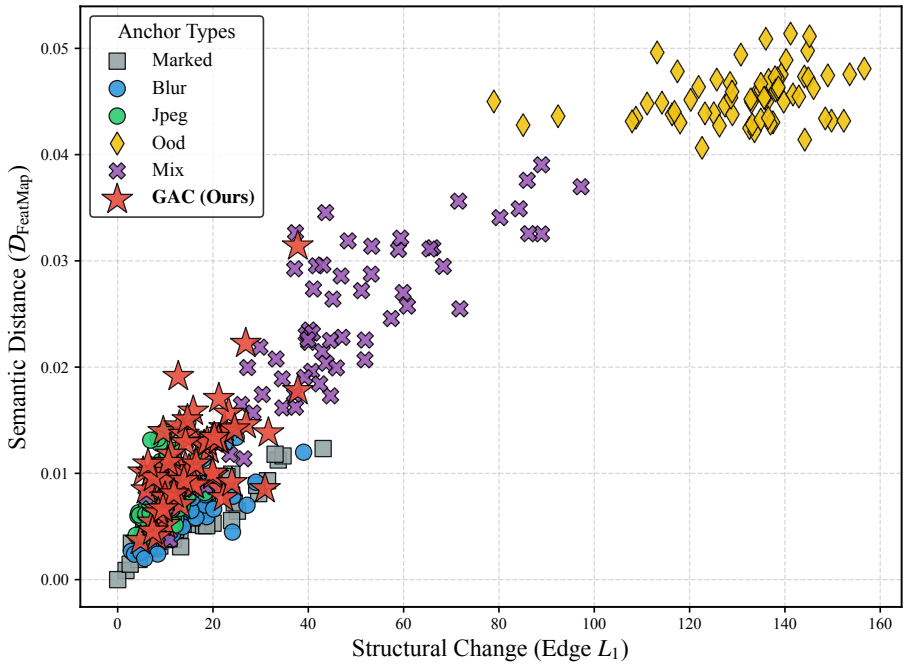


Figure 4: **Safety verification via semantic-structural scatter plot.** The plot contrasts the behavior of GAC ($\star$) against benign and malign anchors. Note that GAC clusters tightly within the low-error **Safe Zone (bottom-left)**, effectively diverging from the structural corruption observed in the **Danger Zone (top-right)** occupied by Mix anchors.

1. Semantic Preservation ($D_{FeatMap}$). Analysis of the semantic distance reveals that the Malign Anchor (Mix), which artificially corrupts defect identity, recorded a high distance of 0.0230, indicating a semantic disconnection from the ground truth. In contrast, GAC recorded 0.0112, showing a preservation capability approximately twice as effective as the Mix anchor. The key finding is that GAC's metric aligns closely with the 'Benign Deformation' group, such as JPEG (0.0082) and Blur (0.0060). This suggests that the instruction-guided editing of GAC acts as a **Safe Generation** process, maintaining the semantic

essence of the defect while removing artifacts.

2. Structural Integrity (Edge L_1 & SSIM). Structural analysis further confirms the practical viability of GAC. The Occluded Baseline (C_{mark}) exhibits an Edge L_1 error of 14.48 due to the presence of marking lines. GAC records a slightly higher error of 15.77; however, this increase is attributed to the *intended removal* of these marking edges. Crucially, this value is distinct from the structural destruction seen in the Mix anchor (45.44). Furthermore, GAC achieves a near-perfect SSIM of 0.9954, significantly outperforming the Malign anchor (0.9344). This proves that GAC faithfully restores the geometric shape of the defect hidden beneath the markings, rather than hallucinating arbitrary textures.

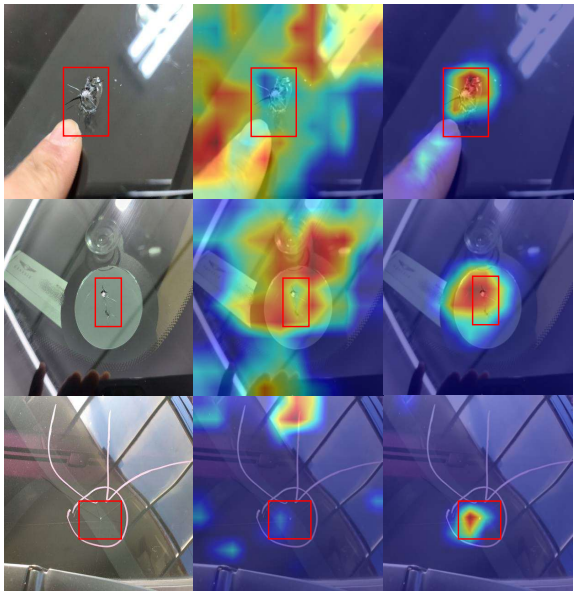
3. Safety Zone Verification. The scatter plot in Fig. 4 visually corroborates the quantitative metrics. On the plane defined by Structural Change (X) and Semantic Distance (Y), the Malign anchor diverges into the top-right **'Danger Zone,'** indicating the simultaneous destruction of both structure and semantics. Conversely, GAC clusters tightly within the bottom-left **'Safe Zone'** alongside Benign Anchors. In conclusion, GAC successfully resolves the trade-off between **Removal** (physically eliminating markings) and **Preservation** (retaining intrinsic defect identity). This validates GAC as a highly efficient, label-free preprocessing tool suitable for industrial pipelines where data integrity is paramount.

5.2. Qualitative Analysis: Attention Rectification via Grad-CAM

This section validates Hypothesis **H1**: *"Training on focused regions rectifies the model's visual attention even when presented with complex scenes."* To rigorously test this, we conducted a cross-domain evaluation using Grad-CAM Selvaraju et al. (2017).

Experimental Setup: The Scale Invariance Stress Test. We trained two separate classifiers (MobileNet v2 backbone): (1) a **Baseline Model** trained

on original full images ($\mathcal{D}_{orig}$), and (2) a **YOLO-Crop Model** trained exclusively on cropped damage regions ($\mathcal{D}_{crop}$). **Crucially, during inference, we fed identical raw full-scale images to both models.** This setup imposes a severe stress test: since the YOLO-Crop model observed only zoomed-in crops during training, testing it on full images introduces a significant **scale discrepancy**. Successful localization under these conditions would prove that the model has learned robust, invariant texture features rather than relying on background context.



(a) Raw Input (b) Baseline (c) YOLO-Crop

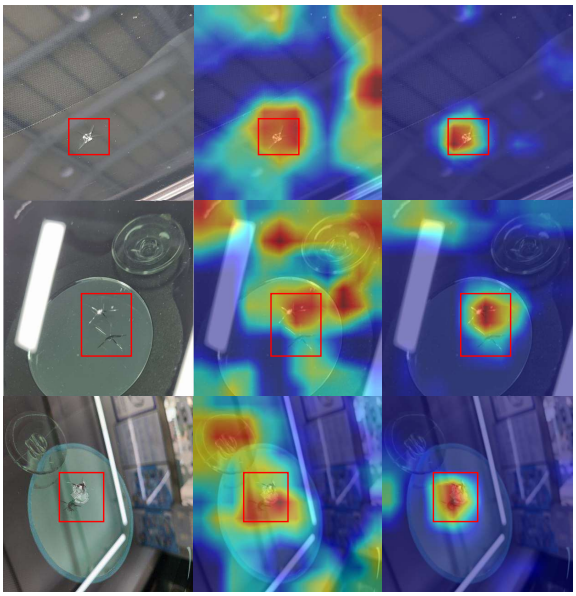
Figure 5: **Visual evidence of shortcut learning vs. rectification.** (a) Raw inputs with spurious artifacts. (b) The Baseline fails to generalize, exhibiting the **Clever Hans** effect by focusing on fingers or lines. (c) Conversely, the proposed method rectifies attention onto the actual damage, validating Hypothesis **H1.**

1. Rectification in Biased Scenarios (Fig. 5). Fig. 5 presents the comparative results on raw images containing markings.

- **Baseline Failure (Contextual Shortcuts):** Despite being trained on full images (domain match), the baseline model (b) fails to focus on the

defect. Its attention is scattered across spurious cues—fingers, reflections, or artificial markings. This confirms that the model minimizes loss by exploiting **contextual shortcuts**.

- **YOLO-Crop Success (Object-Centricity):** Remarkably, even under the scale mismatch condition, our model (c) ignores the background and concentrates its activation precisely on the damage region (Red Box). This demonstrates that geometric focusing fundamentally alters *what* the model learns—shifting from scene-dependent memorization to **object-centric recognition**.



(a) Raw Input (b) Baseline (c) YOLO-Crop

Figure 6: **Validation of background suppression efficacy.** Comparison on clean images reveals that background clutter acts as an independent bias factor. The Baseline (b) is hijacked by high-frequency reflections, whereas the proposed geometric focusing (c) enforces consistent attention alignment with the defect.

2. Robustness in Bias-Free Scenarios (Fig. 6). We extended the analysis to the GAC dataset to determine if background removal itself aids attention,

independent of markings. As shown in Fig. 6, even in clean images, the baseline model exhibits an **Attentional Drift** phenomenon, where focus is hijacked by high-frequency background structures or lighting reflections. Conversely, the YOLO-Crop model consistently **pinpoints** the damage region. This demonstrates that **background clutter** is an independent bias factor clouding the model's judgment. Consequently, YOLO-Crop functions beyond simple cropping; it acts as a **"Data-level Attention Mechanism"** that forcibly corrects the model's cognitive process, simultaneously securing **Interpretability** and **Trustworthiness**.

5.3. Comparative Analysis: Why Geometric Focusing?

In Section 5.2, we visually confirmed that background clutter distracts the model. A natural follow-up question arises: *Is it sufficient to simply remove the background pixels while maintaining the original field of view (FOV)?* Or is *geometric focusing (cropping)* essential?

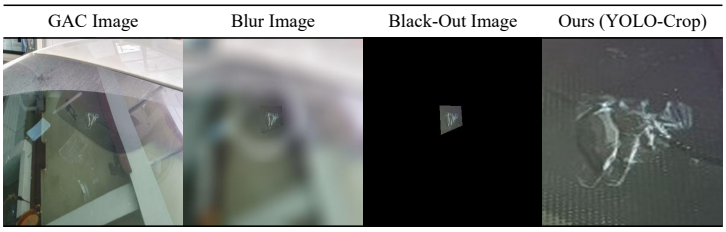


Figure 7: **Overcoming the resolution bottleneck.** Visual comparison between **Passive Suppression** (Blur/Black-Out), which maintains the original FOV, and **Active Geometric Focusing** (YOLO-Crop), which physically amplifies the defect resolution. Note that passive methods fail to solve the low information density problem.

To answer this, we conducted a comparative experiment by deriving four distinct datasets from $\mathcal{D}_{orig}$, applying different background suppression strategies (visualized in Fig. 7). Note that for the passive strategies (Blur and Black-Out), we utilized Ground Truth (GT) coordinates to mask out non-defect regions, creating an *idealized scenario* where both marking artifacts and background noise are perfectly suppressed.

Table 3: **Quantifying the impact of information density.** While passive suppression removes background noise, it hits a **performance ceiling** ($\approx 89\%$) due to the small relative size of defects. In contrast, geometric focusing structurally resolves this bottleneck, yielding a significant accuracy gain.

Dataset	Strategy Characteristics	Accuracy (%)
$\mathcal{D}_{GAC}$	Markings Removed, Background Kept	86.74 ± 0.010
$\mathcal{D}_{blur}$	Background Blurred (Full FOV)	89.38 ± 0.012
$\mathcal{D}_{black}$	Background Blacked-Out (Full FOV)	89.07 ± 0.015
$\mathcal{D}_{crop}$	**Background Removed + Geometric Focusing**	**97.46 ± 0.005**

1. **GAC ($\mathcal{D}_{GAC}$):** Only marking artifacts are removed via generative editing; background is retained.
2. **Gaussian Blur ($\mathcal{D}_{blur}$):** Regions outside the defect GT are heavily blurred ($\sigma = 20$) to destroy high-frequency textures.
3. **Black-Out ($\mathcal{D}_{black}$):** All pixels outside the defect GT are zeroed out ($x_{bg} = 0$), completely eliminating background noise while maintaining the image scale.
4. **YOLO-Crop ($\mathcal{D}_{crop}$):** The damage region is physically cropped and resized to the input resolution (Geometric Focusing).

The classification performance on MobileNet v1 (averaged over 10 runs) is summarized in Table 3.

The results yield three critical insights regarding the nature of industrial visual bias:

1. The Detrimental Role of Background Clutter. Comparing $\mathcal{D}_{GAC}$ (background present) with $\mathcal{D}_{blur/black}$ (background suppressed) reveals a performance gap of approximately 2.5%. Since marking artifacts are suppressed in all three scenarios, this improvement is purely attributable to background removal. This quantitatively supports our claim that background textures act as **confounding bias**, confusing the model even in the absence of spurious markings.

2. The Resolution Bottleneck (Limit of Passive Removal). However, strategies that maintain the full FOV ($\mathcal{D}_{blur}, \mathcal{D}_{black}$) hit a performance ceiling of around 89%. Even with the background completely zeroed out ($\mathcal{D}_{black}$), the model fails to surpass the 90% threshold. This indicates that **"small defect size"** is a bottleneck as critical as bias. In a full-scale image, the damage features are compressed into too few pixels (low information density), limiting the model's discriminative power regardless of background cleanliness.

3. The Necessity of Geometric Focusing. Our implementation, YOLO-Crop, overcomes this limitation, achieving an accuracy of **97.46%**. Unlike passive strategies, YOLO-Crop provides a dual benefit: (1) removing background bias, and (2) **increasing the effective resolution** of the defect via cropping. This implies that for industrial micro-defects, structural cropping is not merely a preprocessing step but a **prerequisite** for high-performance classification, justifying our choice of YOLO over simpler segmentation or masking methods.

5.4. Re-evaluating Performance: The Necessity of the Hybrid Strategy

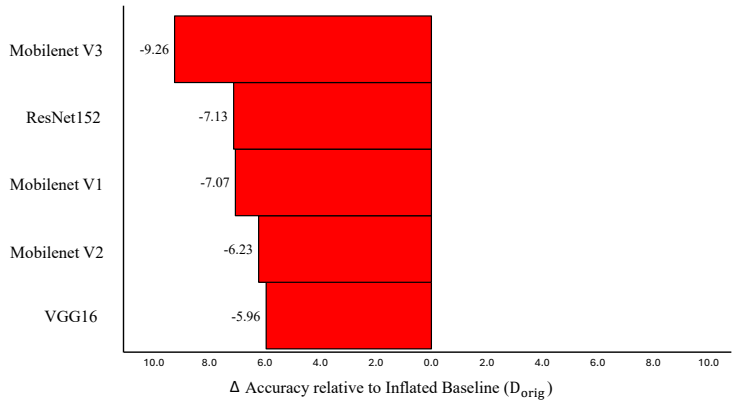


Figure 8: **Quantifying the "Bubble Bursting" effect (D_{orig} vs. D_{GAC}).** The red bars indicate the performance drop observed when spurious markings are removed. This magnitude quantifies the model's reliance on **shortcut learning**, proving that the high accuracy of the baseline was inflated by artifacts.

In this section, we quantitatively analyze the impact of each preprocessing module to justify the proposed Hybrid strategy. We compare the classification performance across three distinct transitions: (1) Marking Removal ($D_{orig} \rightarrow D_{GAC}$), (2) Background Removal in Clean State ($D_{GAC} \rightarrow D_{hybrid}$), and (3) Background Removal in Noisy State ($D_{orig} \rightarrow D_{crop}$).

1. Quantifying Annotation Bias (The Bubble Burst). First, we measure the performance gap between the original dataset (D_{orig}) and the sanitized dataset (D_{GAC}). As shown in Fig. 8, a consistent performance degradation is observed across all five CNN backbones: VGG (-5.96%), ResNet (-7.13%), MobileNet v1 (-7.07%), v2 (-6.23%), and v3 (-9.26%). Since the only variable changed is the removal of markings, this drop quantitatively represents the **magnitude of shortcut learning**. The model on D_{orig} was not recognizing damage but rather "cheating" via markings. Consequently, we define this drop as the **"Bubble Bursting"** effect and establish D_{GAC} as the only legitimate **Fair Baseline**.

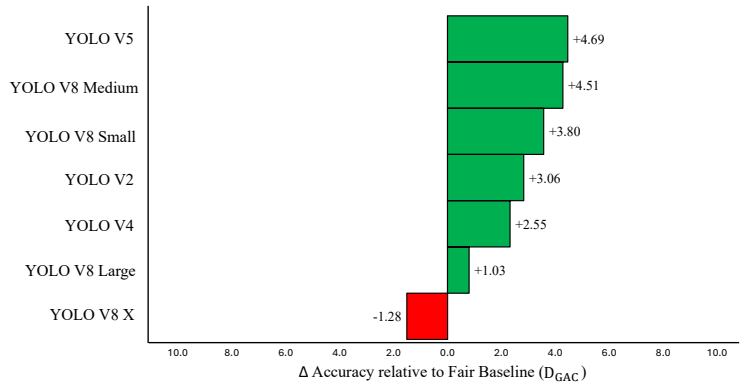


Figure 9: **Isolating the net gain of Geometric Focusing (D_{GAC} vs. D_{hybrid}).** Comparison between the Fair Baseline and the Hybrid Strategy. The consistent improvements (green bars) confirm that background clutter acts as independent noise. By removing it, the model achieves up to **+4.69%** gain in intrinsic information density

2. Isolating the Benefit of Geometric Focusing. With the Fair Baseline (D_{GAC}) established, we evaluate the pure impact of background removal by

comparing it with the Hybrid dataset (D_{hybrid}). Since markings are absent in both, any performance change is strictly attributable to geometric focusing. As illustrated in Fig. 9, applying YOLO-Crop yielded distinct performance gains in 6 out of 7 YOLO variants (+1.03% $\sim$ +4.69%). This quantitative improvement is directly linked to the **Attention Rectification** phenomenon observed in Section 5.2. Just as YOLO-Crop physically blocked background bias and forced the model's attention onto the defect in Grad-CAM visualizations, this attentional concentration translates directly into improved classification accuracy. This proves that removing background clutter is essential for maximizing the model's discriminative power. *(Note: The slight drop in YOLOv8-X is attributed to the excessive capacity of the model over-fitting to minute generative artifacts.)*

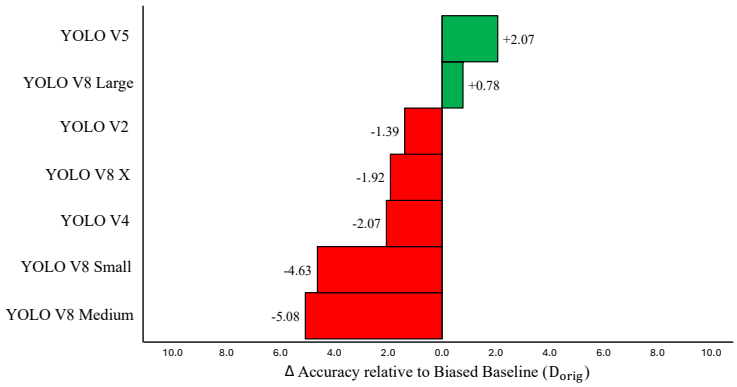


Figure 10: **Instability of naive cropping (D_{orig} vs. D_{crop}).** Unlike the consistent gains in the Hybrid strategy, naive cropping yields fluctuating results (mixed bars). This indicates that **residual internal markings** within the ROI continue to confound the model, creating a "tug-of-war" between intrinsic features and shortcuts

3. The Instability of Naive Cropping (D_{crop}). Finally, we analyze why the Hybrid approach is superior to simply cropping the original images ($D_{orig} \rightarrow D_{crop}$). The results in Fig. 10 reveal a chaotic landscape: while some models show marginal gains, others suffer significant degradation (e.g., YOLOv8 Medium -5.08%). This **Performance Instability** arises from incomplete bias

mitigation. While geometric focusing effectively filters out **external markings** (partial shortcut removal), it is not a complete solution. As will be quantitatively verified in Section 5.5 (Table 4), the ROI still contains **internal residual markings**. Consequently, the model struggles in a "tug-of-war" between learning legitimate defect features and exploiting these remaining internal shortcuts.

Conclusion: The Hybrid strategy (D_{hybrid}) is the only configuration that guarantees **consistent robustness** by creating a fully sanitized environment where both internal shortcuts and background noise are eliminated.

5.5. Quantitative Assessment of Data Reliability ($\mathcal{R}_{data}$)

Table 4: **Quantifying the dual benefits of Quantity and Quality.** The proposed framework addresses industrial data constraints by simultaneously achieving **Implicit Data Expansion** (creating up to 8,029 samples) and **Bias Filtration** (improving reliability from 66% to $\approx$ 90%). Note that D_{GAC} achieves theoretical 100% reliability by removing markings via generative editing.

Dataset / Method	**Total Images**	**Marking-Present Images**	$\mathcal{R}_{data}$ **(%)**
D_{orig}	2,172	739	66.00%
D_{GAC}	2,172	0	**100.00%**
YOLO v2	6,317	1,627	74.00%
YOLO v4	5,985	1,531	75.00%
YOLO v5	2,759	545	80.25%
YOLO v8 Small	8,012	987	87.68%
YOLO v8 Medium	7,715	935	87.88%
YOLO v8 Large	7,626	796	89.56%
YOLO v8 X	**8,029**	819	**89.80%**

This section quantitatively evaluates the impact of YOLO-Crop on the structural reliability and scale of the dataset. Unlike traditional evaluations that focus solely on model *output*, this study adopts a Data-Centric AI perspective to quantify the integrity of the *input* data itself.

To measure dataset trustworthiness, we model the presence of annotation bias as a random variable. Let $\mathcal{D}$ denote the dataset. We define an indicator function $I_{bias}(x)$ for an input sample x, where $I_{bias}(x) = 1$ if spurious marking artifacts (x_{art}) are present within the ROI, and 0 otherwise. Consequently, the bias density (or ratio) $\rho_{bias} \in [0, 1]$ of the dataset is formulated as the expectation of this indicator:

$$\rho_{bias} = \mathbb{E}_{x \sim \mathcal{D}}[I_{bias}(x)] \approx \frac{1}{N} \sum_{i=1}^{N} I_{bias}(x_i) \tag{4}$$

We then define the **Intrinsic Data Reliability ($\mathcal{R}_{data}$)** as the complement of the bias density, expressed as a percentage:

$$\mathcal{R}_{data}(\%) = 100 \cdot (1 - \rho_{bias}) \tag{5}$$

This metric represents the probabilistic proportion of the "Clean Region"—samples free from explicit marking artifacts—within the dataset distribution.

Operational Definition and Interpretation: It is critical to define the scope of $\mathcal{R}_{data}$. Rather than serving as a direct predictor of classification accuracy, this metric functions as a quantitative *"Visual Hygiene Score."* It acts as a diagnostic metric to verify whether the preprocessing pipeline has effectively filtered out explicit bias factors (x_{art}). Ensuring a high $\mathcal{R}_{data}$ establishes a necessary condition for trustworthy learning by minimizing the structural risk of shortcut learning.

Table 4 summarizes the evolution of dataset scale and reliability across various YOLO variants. The results empirically demonstrate that our framework addresses the dual challenges of industrial data: *scarcity* and *bias*.

- **Quantity (Implicit Data Expansion):** YOLO-Crop expands the original 2,172 images to a maximum of 8,029 (YOLO v8 X) by decomposing multiple defect instances within a single image. This yields an **implicit expansion factor of approximately 3.7**$\times$, structurally mitigating the chronic data scarcity in industrial domains.

- **Quality (Bias Filtration):** The original dataset exhibited a low reliability of 66% (i.e., 34% contamination). In contrast, YOLO-Crop improved this to 89.80%, representing a **+23.8%p enhancement** in reliability. This confirms that YOLO functions as an effective "Visual Filter," geometrically excluding markings located outside the defect ROI.

Trade-off between Purity and Information Density: A notable observation is the divergence between reliability and downstream performance. While YOLO v8 X achieved the highest reliability (89.80%), the optimal classification accuracy was recorded with YOLO v5 (Reliability 80.25%, Accuracy 97.46%; see Table A.6). This suggests a critical trade-off: excessively strict cropping (maximizing $\mathcal{R}_{data}$) can induce a Resolution Mismatch. When extremely small defect crops are upscaled to the CNN input size (e.g., 224×224), it results in severe blurring and texture degradation, leading to a loss of high-frequency information essential for classification. Therefore, maintaining a marginal spatial buffer is necessary not for context, but for preserving structural integrity during resizing.

Conclusion and Motivation for Hybrid Strategy: In summary, YOLO-Crop operates as a powerful **Dual-Optimization Mechanism**, physically scaling data volume while statistically enhancing intrinsic reliability. However, it is imperative to acknowledge that a reliability of 89.80% implies that approximately 10.2% of the data still contains artifacts located *inside* the ROI (x_{art}^{int}). Since geometric cropping cannot physically remove these internal biases, the risk of residual shortcut learning persists.

This structural limitation underscores the necessity of our proposed **Hybrid Strategy**. To achieve complete trustworthiness (i.e., $100\%\mathcal{R}_{data}$), a synergistic approach is required: utilizing YOLO-Crop for *macroscopic* spatial focusing to eliminate background noise, while deploying **Generative Annotation Cleaning (GAC)** for *microscopic* pixel-level sanitization to remove internal artifacts. This logical progression justifies the integration of GAC as an essential component for realizing the ultimate "Crop for Trust" framework.

Table 5: **The paradox of high objectness and low separability.** Despite successfully cropping defects (high objectness), end-to-end YOLO models struggle with texture-based classification (low mAP). This **Task Interference** necessitates decoupling localization from semantic classification to resolve the resolution mismatch.

Model	**Precision**	**Recall**	**mAP50**
YOLO v2	0.660	0.530	0.340
YOLO v4	0.740	0.700	0.511
YOLO v5	0.596	0.577	0.556
YOLO v8 Small	0.585	0.557	0.510
YOLO v8 Medium	0.709	0.553	0.562
YOLO v8 Large	**0.769**	0.495	0.521
YOLO v8 X	0.691	0.539	0.542

This section elucidates the design rationale for utilizing YOLO exclusively as a *Spatial Filter* (Cropper) within the preprocessing stage, rather than deploying it as an end-to-end classifier. This architectural decision is not an arbitrary engineering choice but a consequence of **Strategic Decoupling**, necessitated by the inherent trade-offs between one-stage detectors and the structural constraints of industrial data.

The primary bottleneck is the absolute scarcity of domain-specific data ($\sim$2,000 images). One-stage detectors like YOLO must simultaneously optimize dual objective functions: bounding box regression (L_{loc}) and semantic classification (L_{cls}). Training these competing objectives on a limited dataset often induces **Task Interference**, where the model struggles to balance localization precision with semantic discrimination.

Table 5 illustrates the performance metrics when fine-tuning YOLO directly as a multi-class detector. A critical **Performance Imbalance** is observed, which validates our decoupled architecture:

- **High Objectness vs. Low Class Separability:** It is crucial to interpret the 'Recall' values (e.g., 0.495 for YOLO v8 Large) correctly. This metric represents *Class-Specific Recall*, implying the model often confuses 'Repair' with 'Replace'. However, this does not indicate a failure in localization. As verified in Section 5.5, the high reliability score ($\mathcal{R}_{data} \approx 89.8\%$) confirms that the *Objectness Recall*—the ability to locate the existence of a defect regardless of its class—is robust. In our framework, we leverage this strength by using YOLO in a class-agnostic manner.

- **Semantic Bottleneck:** Conversely, the suboptimal mAP50 (0.34–0.56) indicates that while the model successfully localizes the Region of Interest (ROI), it fails to capture the subtle *semantic nuances* required for fine-grained classification.

We identify two structural root causes for this imbalance:

1. **Inductive Bias (Texture vs. Shape):** YOLO architectures are generally optimized for shape-based object recognition (e.g., COCO objects). However, windshield defects are characterized by minute *textural* patterns rather than distinct geometric shapes, which poses a challenge for standard detection backbones.
2. **Resolution Mismatch and Feature Collapse:** In an end-to-end YOLO inference, a small defect region is processed within a minimal feature grid (e.g., 13×13), leading to significant information loss or "Feature Collapse." By decoupling the process, the downstream CNN receives the cropped image resized to a high resolution (e.g., 224×224). This **Input Space Interpolation** preserves the intricate high-frequency details necessary for the model to distinguish between complex damage types.

Accordingly, our pipeline extracts only the **Localization Robustness** of YOLO, while delegating **Semantic Discrimination** to a separate, texture-specialized CNN backbone. Through this **Complementary Integration**, we

effectively resolve the optimization conflict, securing both precise localization and high-fidelity classification capability that outperforms monolithic detection models.

5.7. Ablation Study: Decoupling the Impact of Dual Strategies

Table 6: **Quantifying the trade-off between accuracy and reliability.** The performance gap between $\mathcal{D}_{crop}$ and $\mathcal{D}_{hybrid}$ (4.06%) represents the **"Residual Shortcut Effect"** driven by internal markings. The Hybrid approach sacrifices this spurious gain to secure **complete trustworthiness** and robust generalization.

Dataset	$\mathcal{R}_{data}$	**GAC**	**YOLO-Crop**	**Best Acc**
$\mathcal{D}_{orig}$	66%	✗	✗	95.39%
$\mathcal{D}_{GAC}$	**100%**	✓	✗	88.71%
$\mathcal{D}_{crop}$	89.80%	✗	✓	97.46%
$\mathcal{D}_{hybrid}$	**100%**	✓	✓	**93.40%**

This section dissects the individual and synergistic contributions of the proposed framework's core components: **Visual Sanitization (GAC)** and **Spatial Focusing (YOLO-Crop)**. Table 6 summarizes the trade-offs between Data Reliability ($\mathcal{R}_{data}$) and Best Classification Accuracy across four distinct experimental scenarios.

1. Impact of GAC (Establishing the Truthful Baseline): First, we analyze the transition from $\mathcal{D}_{orig}$ to $\mathcal{D}_{GAC}$. While GAC perfectly restored data reliability to 100%, classification accuracy dropped significantly from 95.39% to 88.71%. As hypothesized in Section 5.4, this reduction is not a degradation of the model's capability but rather the removal of the **"Performance Bubble."** Thus, 88.71% represents the **Truthful Baseline**—the maximum performance achievable using purely intrinsic defect features without spurious aid.

2. Impact of YOLO-Crop (The Hazard of Residual Shortcuts): Second, YOLO-Crop achieved the highest nominal accuracy (97.46%) when applied

alone. By geometrically filtering out background noise ($\mathcal{R}_{data} \rightarrow 89.8\%$), it lowered classification difficulty and maximized learning efficiency. However, calling this the "optimal" strategy is misleading. Although the remaining markings constitute a minor portion (10.2%) of the dataset, their high visual prominence (i.e., strong contrast compared to subtle defects) can disproportionately influence the model's decision boundary. Consequently, the model likely exploits these persistent artifacts as high-confidence cues to inflate the nominal accuracy. This configuration represents a **Performance Ceiling with Bias**, where accuracy is inflated by residual artifacts.

3. The Hybrid Strategy (Synergistic Reliability): Finally, the Hybrid scenario ($\mathcal{D}_{hybrid}$) combines both techniques, maintaining 100% data reliability while achieving an accuracy of 93.40%. This is a substantial **+4.69%** improvement over the GAC-only truthful baseline.

Crucially, we quantify the **"Residual Shortcut Effect"** as the performance gap between the biased YOLO-Crop and the sanitized Hybrid model:

$$\Delta_{shortcut} = Acc(\mathcal{D}_{crop}) - Acc(\mathcal{D}_{hybrid}) = 97.46\% - 93.40\% = 4.06\%$$

This 4.06% gap signifies the magnitude of performance gained solely from spurious markings remaining inside the ROI. By consciously sacrificing this spurious gain, the Hybrid model not only achieves **Zero Visual Bias** but also prevents potential generalization failures in real-world deployments where markings may differ or be absent. This establishes it as the most **trustworthy solution.**

Conclusion: In summary, GAC functions as an **Integrity Enforcer** ensuring data fairness, while YOLO-Crop acts as a **Focus Amplifier** enhancing intrinsic feature extraction. Their combination provides the only viable path to building an Industrial AI system that is both interpretable and robust against shortcut learning.

6. Conclusion

This study addressed the persistent structural dilemma of data scarcity and visual bias in industrial environments. We identified the phenomenon

of **"Shortcut Learning,"** driven by spurious correlations, and proposed the **"Crop for Trust"** framework to fundamentally mitigate this at the input level.

Our approach focused on restoring data integrity through a dual-branch strategy of **Geometric Focusing (YOLO-Crop)** and **Generative Sanitization (GAC)**. By establishing a **"Truthful Baseline"** via GAC, we empirically demonstrated that the state-of-the-art performance metrics were largely inflated by spurious correlations—a phenomenon we termed the **"Performance Bubble."** Furthermore, while simple geometric focusing (YOLO-Crop) recovered this drop, our proposed **Hybrid Strategy** went a step further. It achieved a genuine accuracy improvement of **+4.69%** over the baseline while ensuring **Zero Visual Bias**, proving itself as a robust **Pre-hoc Debiasing** mechanism.

The core contributions of this study are summarized as follows:

1. **Unmasking and Quantifying Bias:** We proved that data-level visual bias distorts the model's *Attention Distribution*. Crucially, we quantified the **"Residual Shortcut Effect"** ($\Delta_{shortcut} \approx 4.06\%$), revealing that even cropped images can induce shortcut learning if internal artifacts are not sanitized.
2. **Reliability-Aware Methodology:** We introduced a novel probabilistic metric, **Intrinsic Data Reliability ($\mathcal{R}_{data}$)**, and demonstrated that our framework improves this reliability by up to **23.8%**. Additionally, the framework provides an **implicit data expansion of 3.7**$\times$, simultaneously resolving scarcity and bias.
3. **Strategic Decoupling for Data-Centric AI:** By decoupling localization (YOLO) from semantic classification (CNN), we resolved the optimization conflicts inherent in monolithic detectors. This paradigm shift secures both **Interpretability** and **Robustness** without requiring complex architectural modifications.

Limitations and Future Work: While the GAC module effectively restores visual fairness, the inference latency of generative models remains a constraint for real-time edge deployment. Future research will focus on **Knowledge**

Distillation, transferring the debiasing capability of GAC into lightweight filtering modules. Ultimately, 'Crop for Trust' offers a practical, scalable solution to overcome the limitations of "in-the-wild" industrial data, serving as a cornerstone for building **Trustworthy Industrial Vision Systems**.

Data and Code Availability Statement

The source code used in this study is publicly available at the GitHub repository linked in the abstract. The industrial dataset used in this study is confidential and available from the corresponding author upon reasonable request.

Data and Code Availability Statement

The code supporting the findings of this study is openly available at https://huggingface.co/spaces/Crop-Trust-Research/Crop-for-Trust-Demo, and the data are available from the corresponding author, SeungHyeok Hong (shongdr@gmail.com), upon reasonable request.

References

Aldubaikhi, A., & Patel, S. (2025). Advancements in small-object detection: Approaches, datasets, benchmarks, and practical guidance. *Applied Sciences*, *15*, 11882.

Birhane, A., & Prabhu, V. U. (2021). Large image datasets: A pyrrhic win for computer vision? In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*.

Buongiorno, D., Brunetti, A., & Longo, N. (2023). Deep learning for automatic vision-based recognition of industrial surface defects: A survey. *IEEE Access*, *11*, 10113226.

Corbin, A., & Marques, O. (2023). Assessing bias in skin lesion classifiers with contemporary deep learning and post-hoc explainability techniques. *IEEE Access*, *11*, 10162202.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*.

Elsharif, W., Alzubaidi, M., & Agus, M. (2025). Cultural bias in text-to-image models: A systematic review of bias identification, evaluation, and mitigation strategies. *IEEE Access*, *13*, 11071263.

Gill, N., Hall, P., Montgomery, K., & Schmidt, N. (2020). A responsible machine learning workflow with focus on interpretable models, post-hoc explanation, and discrimination testing. *Information*, *11*, 137.

Google DeepMind (2025). Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint*, . arXiv:2507.06261.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 770–778). IEEE. doi:10.1109/CVPR.2016.90.

Liang, P., Zhao, Z., Lin, J., & Wang, X. (2023). Redefining background bias in visual recognition: When and why context helps. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Lugmayr, A., Danelljan, M., Romero, A., Yu, F., Timofte, R., & Van Gool, L. (2022). Repaint: Inpainting using denoising diffusion probabilistic models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 11461–11471).

Man, Y., Li, Y., & Ma, L. (2024). Clipaway: Harmonizing focused embeddings for removal of objects via diffusion inpainting. *arXiv preprint*, . arXiv:2406.09368.

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You only look once: Unified, real-time object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 779–788). IEEE. doi:10.1109/CVPR.2016.91.

Sagawa, S., Koh, P. W., Hashimoto, T. B., & Liang, P. (2020). Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations (ICLR)*.

Sampath, V., Maurtua, I., Aguilar-Martin, J. J., & Gutierrez, A. (2021). A survey on generative adversarial networks for imbalance problems in computer vision tasks. *Journal of Big Data*, *8*, 1–37.

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)* (pp. 618–626). IEEE.

Tan, M., & Le, Q. V. (2019). Efficientnet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th International Conference on Machine Learning (ICML)* (pp. 6105–6114).

Wu, P., He, X., Dai, W., Zhou, J., Shang, Y., & Fan, Y. (2025). A review on research and application of ai-based image analysis in the field of computer vision. *IEEE Access*, *13*, 10979847.

Yun, S., Han, D., Oh, S. J., Chun, S., Choe, J., & Yoo, Y. (2019). Cutmix: Regularization strategy to train strong classifiers with localizable features. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)* (pp. 6023–6032).

Zeng, B., Yin, Y., & Liu, Z. (2024). Understanding bias in large-scale visual datasets. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Zhang, H., Cisse, M., Dauphin, Y. N., & Lopez-Paz, D. (2018). mixup: Beyond empirical risk minimization. In *International Conference on Learning Representations (ICLR)*.

Appendix A. Detailed Performance Analysis across CNN Architectures

In this appendix, we provide the comprehensive numerical results of the classification performance evaluation discussed in Section 5.4. To verify the **architecture-agnostic** effectiveness of the proposed framework, experiments were conducted on five distinct CNN backbones: VGG16, ResNet152, and MobileNet (v1, v2, v3). Ensuring statistical reliability, all experiments were performed using **10-fold independent runs** under random seed initialization. The results are reported as Mean Accuracy (%) ± Standard Deviation.

Formatting Convention: To facilitate comparative analysis, Tables A.7 and A.8 adhere to the following conventions:

- **Bold Values:** Indicate the highest accuracy achieved among the five CNN backbones for a specific dataset configuration (Row-wise Maximum).
- **Green Values:** Denote a performance improvement compared to the respective Baseline model.
- **Red Values:** Indicate a performance decline compared to the Baseline.

Appendix A.1. Performance on Biased Scenarios (Original Baseline)

Table A.7 details the performance shifts when applying YOLO-Crop to the biased Original Dataset ($\mathcal{D}_{orig}$). These results correspond to the summary visualization in Figure 10 of the main text.

- **Accuracy Gain:** Applying YOLO v5-based cropping to the MobileNet v1 backbone achieved a maximum accuracy of **97.46%**. This represents a substantial improvement of **+3.65%p** compared to the baseline accuracy of 93.81%.

Table A.7: **Numerical breakdown of biased scenarios (D_{orig} vs. D_{crop}).** This table provides the quantitative basis for the "Instability" analysis in Fig 10. While YOLO v5-Crop achieves the peak accuracy (97.46%), other variants exhibit high variance, confirming the risks of naive cropping without internal sanitization.

Dataset	VGG16	ResNet152	MobileNet v1	MobileNet v2	MobileNet v3
Original ($\mathcal{D}_{orig}$)	94.67 ± 1.44	94.93 ± 1.87	93.81 ± 0.65	91.15 ± 1.44	$\mathbf{95.39 \pm 0.56}$
YOLO v2-Crop	92.52 ± 1.50	92.08 ± 0.30	92.66 ± 0.61	91.43 ± 0.20	$\mathbf{94.00 \pm 0.33}$
YOLO v4-Crop	91.95 ± 0.80	91.42 ± 0.54	92.92 ± 0.98	91.75 ± 0.28	$\mathbf{93.32 \pm 0.54}$
YOLO v5-Crop	97.26 ± 1.26	96.39 ± 0.50	$\mathbf{97.46 \pm 0.57}$	94.11 ± 0.40	96.07 ± 0.39
YOLO v8 S-Crop	$\mathbf{90.76 \pm 1.03}$	89.53 ± 0.72	88.11 ± 0.48	90.15 ± 1.13	88.97 ± 0.42
YOLO v8 M-Crop	88.70 ± 0.75	$\mathbf{90.31 \pm 0.80}$	89.44 ± 1.00	89.68 ± 0.58	89.50 ± 0.70
YOLO v8 L-Crop	93.87 ± 1.70	94.52 ± 0.42	94.30 ± 0.63	93.45 ± 0.53	$\mathbf{96.17 \pm 0.62}$
YOLO v8 X-Crop	93.37 ± 0.90	92.72 ± 0.64	92.53 ± 0.71	91.78 ± 0.92	$\mathbf{93.47 \pm 0.50}$

- **Stability Improvement:** A notable observation is the reduction in variance. The MobileNet v2 model exhibited a standard deviation of 1.439 on the original dataset, which drastically decreased to **0.403** upon applying YOLO v5-Crop. This indicates that YOLO-Crop mitigates training instability caused by background noise, enhancing **Convergence Consistency**.

Appendix A.2. Performance on Bias-Free Scenarios (Fair Baseline)

Table A.8 presents the experimental results based on the sanitized GAC dataset ($\mathcal{D}_{GAC}$). These results correspond to the summary visualization in Figure 9.

- **Robustness Verification:** Even in a fair environment where bias is removed, the Hybrid strategy ($\mathcal{D}_{hybrid}$) demonstrated consistent performance improvements. Specifically, the combination of YOLO v5 and VGG16 recorded an accuracy of **93.40%**, marking an improvement of **+4.69%p** over the truthful GAC baseline (88.71%).

- **Architectural Consistency:** Performance gains were consistent across both lightweight (MobileNet series) and high-capacity (ResNet, VGG)

Table A.8: **Validation of the Hybrid Strategy (D_{GAC} vs. D_{hybrid}).** Corresponding to Fig 9, these results confirm that the proposed Hybrid approach consistently outperforms the Fair Baseline (D_{GAC}) across diverse CNN architectures. This empirically proves the architecture-agnostic robustness of removing background clutter from sanitized images.

Dataset	**VGG16**	**ResNet152**	**MobileNet v1**	**MobileNet v2**	**MobileNet v3**
Fair Baseline ($\mathcal{D}_{GAC}$)	$\mathbf{88.71 \pm 1.53}$	87.80 ± 1.04	86.74 ± 1.03	84.92 ± 0.98	86.13 ± 0.96
YOLO v2-Hybrid	89.79 ± 0.92	$\mathbf{91.77 \pm 0.50}$	91.11 ± 0.52	89.30 ± 3.05	88.22 ± 1.20
YOLO v4-Hybrid	90.28 ± 1.74	90.33 ± 0.79	90.93 ± 0.36	$\mathbf{91.26 \pm 1.11}$	89.06 ± 0.87
YOLO v5-Hybrid	$\mathbf{93.40 \pm 0.96}$	91.57 ± 0.88	92.53 ± 0.68	91.87 ± 0.72	90.50 ± 0.65
YOLO v8 S-Hybrid	91.84 ± 1.33	$\mathbf{92.51 \pm 0.67}$	91.95 ± 0.71	92.06 ± 0.93	88.43 ± 0.97
YOLO v8 M-Hybrid	90.67 ± 1.62	$\mathbf{93.22 \pm 1.35}$	93.12 ± 1.30	92.44 ± 1.08	91.80 ± 1.30
YOLO v8 L-Hybrid	87.58 ± 1.25	$\mathbf{89.74 \pm 1.12}$	89.23 ± 0.72	85.15 ± 1.40	88.34 ± 0.64
YOLO v8 X-Hybrid	85.54 ± 1.79	$\mathbf{87.43 \pm 1.21}$	86.75 ± 1.06	82.97 ± 1.50	86.21 ± 1.71

models. While slight fluctuations were observed in YOLO v8-X, this is likely attributed to the **model's excessive capacity being oversensitive to subtle distributional shifts** inherent in the generative restoration process. Despite this outlier, the majority of combinations confirmed simultaneous improvements in both trustworthiness and accuracy.